

## Ex.No.6 INTERFACING STEPPER MOTOR WITH 8051

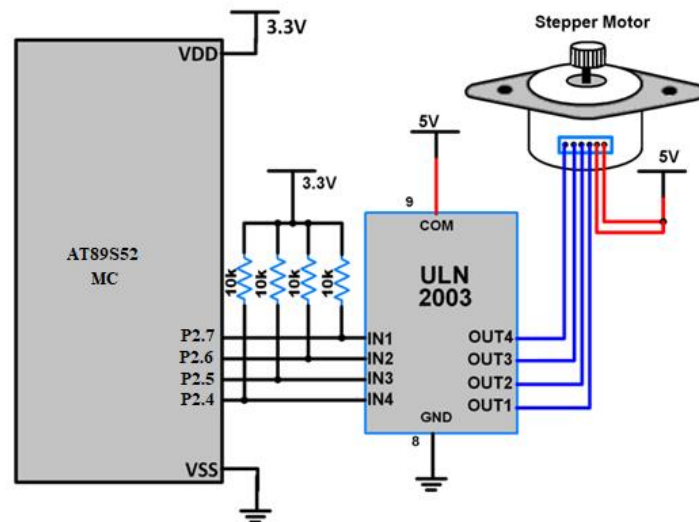
### AIM

To interface the stepper motor and control the clock wise and counter clock wise rotation using embedded c program with 8051 Microcontroller.

### EQUIPMENT

Keil uvision-IDE, 8051 development board, FRC cables, Stepper motor.

### CIRCUIT DIAGRAM



### PROCEDURE

**Step 1:** Give a double click on  $\mu$ vision 5 icon.

**Step 2:** To create a new project, click on the project window and select new  $\mu$ vision project. Create a new folder with a project name to save the project.

**Step 3:** Select the target microcontroller i.e. 8051AH from Intel and click OK

**Step 4:** Click YES to Copy STARTUP.A51 to Project Folder and Add file to Project. Type the Embedded C Program and save the file in the folder created with c extension.

**Step 5:** Add the .c File to source group by right click on Source Group, choose “**ADD files to Group**” option.

**Step 6:** Right click on target in the project window and select “**Options for Target**”

**Step 7:** Go to “**OUTPUT**” option in the popped window and check (✓) “**Create Hex File**”.

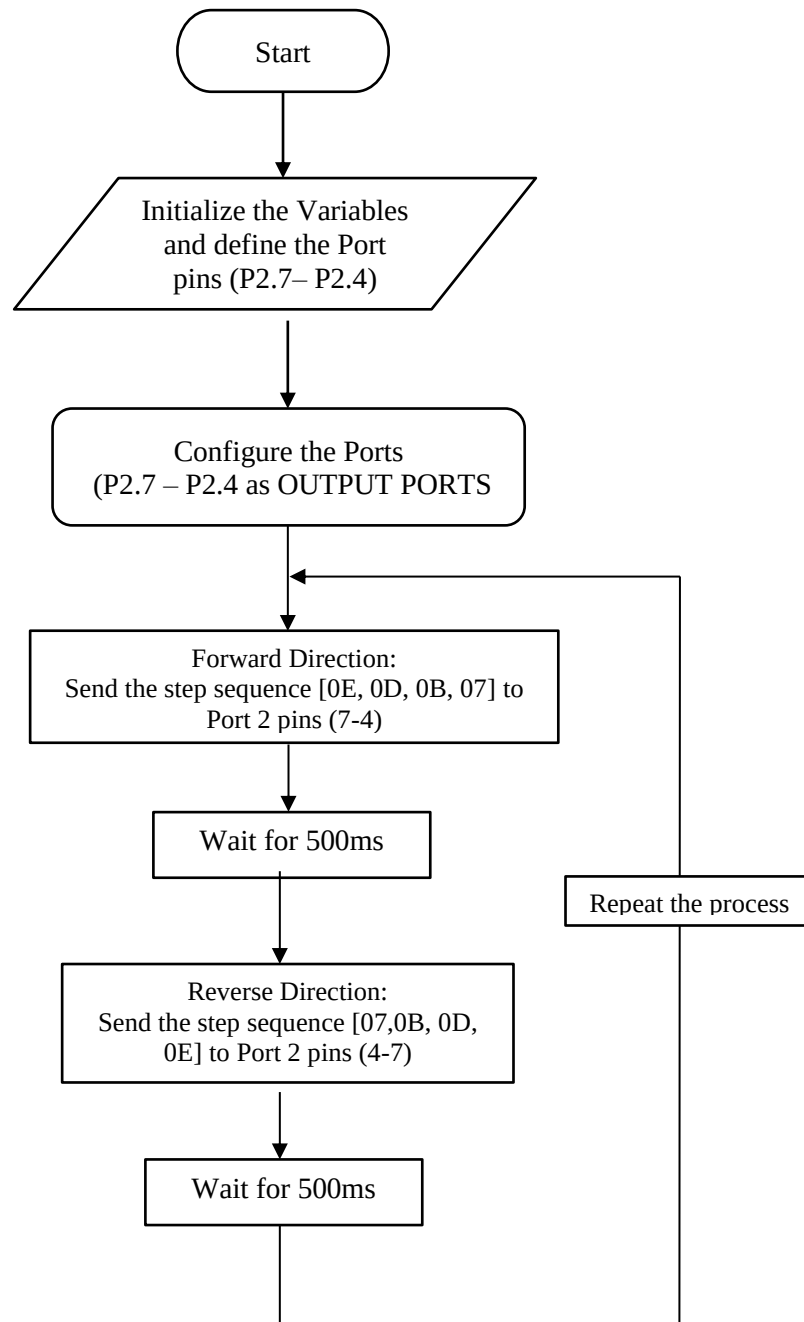
**Step 8:** In the same window go to Linker option and check (✓) “**Use Memory Layout from Target Dialog**” and click OK.

**Step 9:** To compile the project, click Project and select Build Target option or press F7.

**Step 10:** Next open Progisp Software, select the microcontroller (AT89S52), enable “verify signature”, “Chip Erase”, “program FLASH” and “verify FLASH”. Click Load Flash to burn the code (hex file) into the controller. Click on the AUTO button to upload the code.

**Step 11:** Check the output of stepper motor for clockwise and anti-clockwise rotation.

## FLOWCHART



## MOTOR DRIVE MODE

| P2.4 | P2.5 | P2.6 | P2.7 | HEX |
|------|------|------|------|-----|
| D    | C    | B    | A    |     |
| 1    | 1    | 1    | 0    | 0E  |
| 1    | 1    | 0    | 1    | 0D  |
| 1    | 0    | 1    | 1    | 0B  |
| 0    | 1    | 1    | 1    | 07  |

**ALGORITHM:**

1. Initialize Port 2 pins (P2.7, P2.6, P2.5, P2.4)
2. Configure Port 2 pins P2.7, P2.6, P2.5, and P2.4 as output port pins.
3. To rotate the stepper motor in the forward direction, send the hex values (0E, 0D, 0B, 07) to the Port 2 pins with a delay
4. Repeat step 3 to rotate the motor continuously in the forward direction.
5. To rotate the stepper motor in the reverse direction, send the hex values (07, 0B, 0D, 0E) to the Port 2 pins with a delay.
6. Repeat step 5 to rotate the motor continuously in the reverse direction.

**PROGRAM****RESULT**

Thus, interfacing of stepper motor is done and verified using Embedded C program with 8051 Development KIT.